

DBMS – PRODUCT AND CATEGORY MANAGEMENT

Project Title: Helios Watch Store – E-Commerce Watch Management System

Skill Task: DBMS

Week: 2

1. Database Design

1.1 Category Table

Attribute	Data Type	Constraint
Category_ID	NUMBER	Primary Key
Category_Name	VARCHAR2(50)	NOT NULL
Description	VARCHAR2(255)	-

1.2 Product Table

Attribute	Data Type	Constraint
Product_ID	NUMBER	Primary Key
Product_Name	VARCHAR2(100)	NOT NULL
Brand_ID	NUMBER	—
Category_ID	NUMBER	NOT NULL, Foreign Key
Price	NUMBER (10,2)	NOT NULL
Colors	VARCHAR2(100)	—
Strap_Material	VARCHAR2(50)	—
Dial_Color	VARCHAR2(50)	—
Water_Resistance	VARCHAR2(50)	—
Stock	NUMBER	NOT NULL

2. Primary and Foreign Key Relationship

Category → Product

- Category.Category_ID → Primary Key
- Product.Product_ID → Primary Key
- Product.Category_ID → Foreign Key referencing Category.Category_ID
- Relationship: **One-to-Many (1:M)**

3. Table Creation

3.1 Create Category Table

```
SQL> CREATE TABLE Category (  
2     Category_ID    NUMBER PRIMARY KEY,  
3     Category_Name  VARCHAR2(50) NOT NULL,  
4     Description     VARCHAR2(255)  
5 );
```

Table created.

3.2 Create Product Table

```
CREATE TABLE Product (  
2     Product_ID      NUMBER PRIMARY KEY,  
3     Product_Name     VARCHAR2(100) NOT NULL,  
4     Brand_ID         NUMBER,  
5     Category_ID      NUMBER NOT NULL,  
6     Price            NUMBER(10,2) NOT NULL,  
7     Colors           VARCHAR2(100),  
8     Strap_Material   VARCHAR2(50),  
9     Dial_Color       VARCHAR2(50),  
10    Water_Resistance VARCHAR2(50),  
11    Stock            NUMBER NOT NULL,  
12    CONSTRAINT FK_Product_Category  
13         FOREIGN KEY (Category_ID)  
14         REFERENCES Category(Category_ID)  
15 );
```

Table created.

4. Data Insertion

4.1 Category Records

```
SQL> INSERT INTO Category (Category_ID, Category_Name, Description)
      2 VALUES (1, 'Analog', 'Classic analog watches');

1 row created.

SQL>
SQL> INSERT INTO Category (Category_ID, Category_Name, Description)
      2 VALUES (2, 'Digital', 'Modern digital watches');

1 row created.

SQL>
SQL> INSERT INTO Category (Category_ID, Category_Name, Description)
      2 VALUES (3, 'Smart Watch', 'Smart watches with advanced features');

1 row created.

SQL>
SQL> INSERT INTO Category (Category_ID, Category_Name, Description)
      2 VALUES (4, 'Sports', 'Watches designed for sports and outdoor activities');

1 row created.

SQL>
SQL> INSERT INTO Category (Category_ID, Category_Name, Description)
      2 VALUES (5, 'Luxury', 'Premium and luxury watches');

1 row created.
```

Activate Windows
Go to Settings to activate Windows.

4.2 Product Records

```
SQL> INSERT INTO Product
  2 VALUES (101, 'Classic Silver Watch', 1, 1, 2499.00, 'Silver, Black', 'Leather', 'Black', '50m', 25);

1 row created.

SQL>
SQL> INSERT INTO Product
  2 VALUES (102, 'Elegant Gold Watch', 2, 1, 3999.00, 'Gold', 'Stainless Steel', 'White', '50m', 18);

1 row created.

SQL>
SQL> INSERT INTO Product
  2 VALUES (103, 'Digital Sport Watch', 3, 2, 1999.00, 'Black, Blue', 'Silicone', 'Black', '100m', 30);

1 row created.

SQL>
SQL> INSERT INTO Product
  2 VALUES (104, 'Smart Fit Pro', 4, 3, 5999.00, 'Black, Grey', 'Silicone', 'Black', '50m', 15);

1 row created.

SQL>
SQL> INSERT INTO Product
  2 VALUES (105, 'Smart Active Watch', 5, 3, 7499.00, 'Black, Green', 'Silicone', 'Black', '50m', 12);

1 row created.
```

Activate Windows

```
SQL>
SQL> INSERT INTO Product
  2 VALUES (106, 'Sports Chronograph', 6, 4, 4599.00, 'Black, Red', 'Stainless Steel', 'Black', '100m', 20);

1 row created.

SQL>
SQL> INSERT INTO Product
  2 VALUES (107, 'Outdoor Pro Watch', 7, 4, 5299.00, 'Black, Orange', 'Silicone', 'Black', '200m', 14);

1 row created.

SQL>
SQL> INSERT INTO Product
  2 VALUES (108, 'Royal Gold Edition', 8, 5, 12999.00, 'Gold', 'Stainless Steel', 'Blue', '100m', 8);

1 row created.

SQL>
SQL> INSERT INTO Product
  2 VALUES (109, 'Premium Leather Watch', 9, 5, 8999.00, 'Brown, Black', 'Leather', 'White', '50m', 10);

1 row created.

SQL>
SQL> INSERT INTO Product
  2 VALUES (110, 'Classic Black Watch', 10, 1, 2999.00, 'Black', 'Leather', 'Black', '50m', 22);

1 row created.
```

Activate Windows

Go to Settings to activate Windows.

5. Data Retrieval

5.1 Display Category Records

```
SQL> SELECT * FROM Category;
```

```
CATEGORY_ID CATEGORY_NAME
```

```
-----
```

```
DESCRIPTION
```

```
-----
```

```
1 Analog  
Classic analog watches
```

```
2 Digital  
Modern digital watches
```

```
3 Smart Watch  
Smart watches with advanced features
```

```
CATEGORY_ID CATEGORY_NAME
```

```
-----
```

```
DESCRIPTION
```

```
-----
```

```
4 Sports  
Watches designed for sports and outdoor activities
```

```
5 Luxury  
Premium and luxury watches
```

5.2 Display Product Records

```
SQL> SELECT * FROM PRODUCT;
```

PRODUCT_ID

PRODUCT_NAME

BRAND_ID CATEGORY_ID PRICE

COLORS

STRAP_MATERIAL

DIAL_COLOR

WATER_RESISTANCE STOCK

103

PRODUCT_ID

PRODUCT_NAME

BRAND_ID CATEGORY_ID PRICE

COLORS

STRAP_MATERIAL

DIAL_COLOR

Activate Windows
Go to Settings to activate Windows.

WATER_RESISTANCE STOCK

103

PRODUCT_ID

PRODUCT_NAME

BRAND_ID CATEGORY_ID PRICE

COLORS

STRAP_MATERIAL

DIAL_COLOR

WATER_RESISTANCE STOCK

Digital Sport Watch

PRODUCT_ID

PRODUCT_NAME

BRAND_ID CATEGORY_ID PRICE

COLORS

STRAP_MATERIAL

Activate Windows
Go to Settings to activate Windows.

DIAL_COLOR

WATER_RESISTANCE

STOCK

3

2

1999

PRODUCT_ID

PRODUCT_NAME

BRAND_ID

CATEGORY_ID

PRICE

COLORS

STRAP_MATERIAL

DIAL_COLOR

WATER_RESISTANCE

STOCK

Black, Blue

PRODUCT_ID

PRODUCT_NAME

BRAND_ID

CATEGORY_ID

PRICE

COLORS

Activate Windows

Go to Settings to activate Windows.

```
-----  
STRAP_MATERIAL  
-----  
DIAL_COLOR  
-----  
WATER_RESISTANCE          STOCK  
-----  
Silicone  
  
PRODUCT_ID  
-----  
PRODUCT_NAME  
-----  
  BRAND_ID CATEGORY_ID    PRICE  
-----  
COLORS  
-----  
STRAP_MATERIAL  
-----  
DIAL_COLOR  
-----  
WATER_RESISTANCE          STOCK  
-----  
Black  
  
PRODUCT_ID  
-----  
PRODUCT_NAME  
-----  
  BRAND_ID CATEGORY_ID    PRICE  
-----
```

```
-----  
COLORS  
-----  
STRAP_MATERIAL  
-----  
DIAL_COLOR  
-----  
WATER_RESISTANCE          STOCK  
-----  
50m                        22  
  
PRODUCT_ID  
-----  
PRODUCT_NAME  
-----  
  BRAND_ID CATEGORY_ID    PRICE  
-----  
COLORS  
-----  
STRAP_MATERIAL  
-----  
DIAL_COLOR  
-----  
WATER_RESISTANCE          STOCK  
-----  
  
10 rows selected.  
  
SQL> |
```

6. Product Update

Update the price and stock of **Classic Silver Watch**:

```
SQL> UPDATE Product
  2  SET Price = 2799.00,
  3      Stock = 30
  4  WHERE Product_ID = 101;

1 row updated.
```

Update the price of **Elegant Gold Watch**:

```
SQL> UPDATE Product
  2  SET Price = 4299.00
  3  WHERE Product_Name = 'Elegant Gold Watch';

1 row updated.

SQL> █
```

7. Product Deletion

Delete the product **Classic Black Watch**:

```
SQL> DELETE FROM Product
  2  WHERE Product_ID = 110;

1 row deleted.
```

Delete the product **Premium Leather Watch**:

```
SQL> DELETE FROM Product
      2 WHERE Product_Name = 'Premium Leather Watch';

1 row deleted.
```

8. Category-wise Product Reports

8.1 Product Listing by Category

This report displays the **category name along with the products belonging to each category**, including their price and stock.

```
SQL> SELECT
      2     c.Category_Name,
      3     p.Product_ID,
      4     p.Product_Name,
      5     p.Price,
      6     p.Stock
      7 FROM Category c
      8 JOIN Product p
      9     ON c.Category_ID = p.Category_ID
     10 ORDER BY c.Category_Name, p.Product_Name;
```

CATEGORY_NAME	PRODUCT_ID
PRODUCT_NAME	
PRICE	STOCK
Analog	101
Classic Silver Watch	
2799	30
Analog	102
Elegant Gold Watch	
4299	18

CATEGORY_NAME	PRODUCT_ID
PRODUCT_NAME	

Activat
6 of 6

PRICE		STOCK
Digital		103
Digital Sport Watch		
1999		30
Luxury		108
Royal Gold Edition		
CATEGORY_NAME		PRODUCT_ID
PRODUCT_NAME		
PRICE		STOCK
12999		8
Smart Watch		105
Smart Active Watch		
7499		12
Smart Watch		104
CATEGORY_NAME		PRODUCT_ID
PRODUCT_NAME		
PRICE		STOCK

Smart Watch		104
CATEGORY_NAME		PRODUCT_ID
PRODUCT_NAME		
PRICE		STOCK
Smart Fit Pro		15
5999		
Sports		107
Outdoor Pro Watch		
5299		14
CATEGORY_NAME		PRODUCT_ID
PRODUCT_NAME		
PRICE		STOCK
Sports		106
Sports Chronograph		
4599		20

8 rows selected.

8.2 Products in Smart Watch Category

This report displays the products belonging specifically to the **Smart Watch** category.

```
SQL> SELECT
  2     p.Product_ID,
  3     p.Product_Name,
  4     p.Price,
  5     p.Stock
  6 FROM Product p
  7 JOIN Category c
  8     ON p.Category_ID = c.Category_ID
  9 WHERE c.Category_Name = 'Smart Watch';
```

PRODUCT_ID

PRODUCT_NAME

PRICE

STOCK

104

Smart Fit Pro

5999

15

105

Smart Active Watch

7499

12

PRODUCT_ID

PRODUCT_NAME

PRICE

STOCK

8.3 Product Count by Category

This report displays the **number of products available in each category**

```
SQL> SELECT
  2     c.Category_Name,
  3     COUNT(p.Product_ID) AS Total_Products,
  4     NVL(SUM(p.Stock), 0) AS Total_Stock
  5 FROM Category c
  6 LEFT JOIN Product p
  7     ON c.Category_ID = p.Category_ID
  8 GROUP BY c.Category_Name
  9 ORDER BY c.Category_Name;
```

CATEGORY_NAME	TOTAL_PRODUCTS	TOTAL_STOCK
-----	-----	-----
Analog	2	48
Digital	1	30
Luxury	1	8
Smart Watch	2	27
Sports	2	34